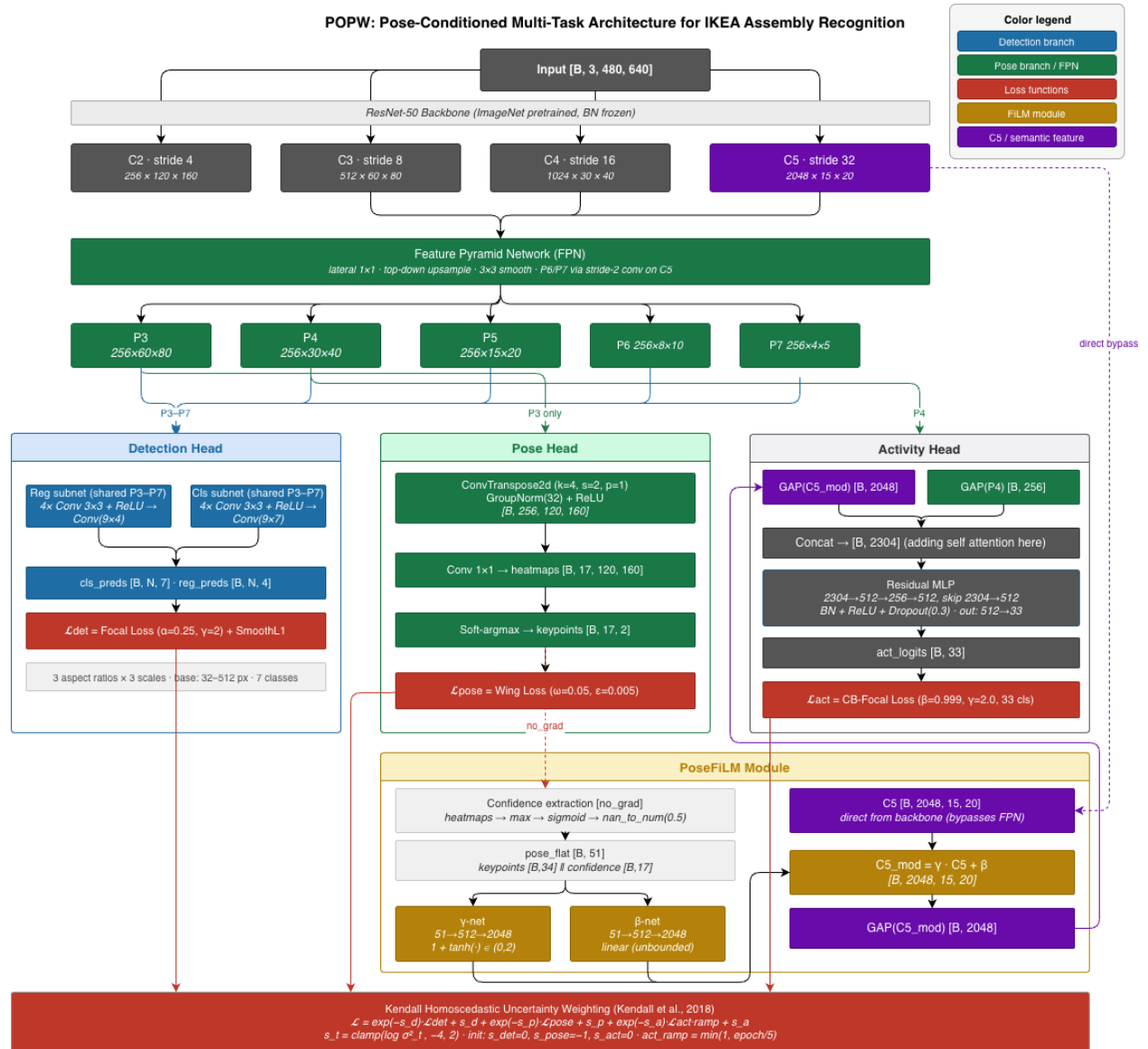


# 14th Meeting 04/09

What was discussed from previous meeting:

- Correct the flow, correct the pose head, add the self attention to the activity head

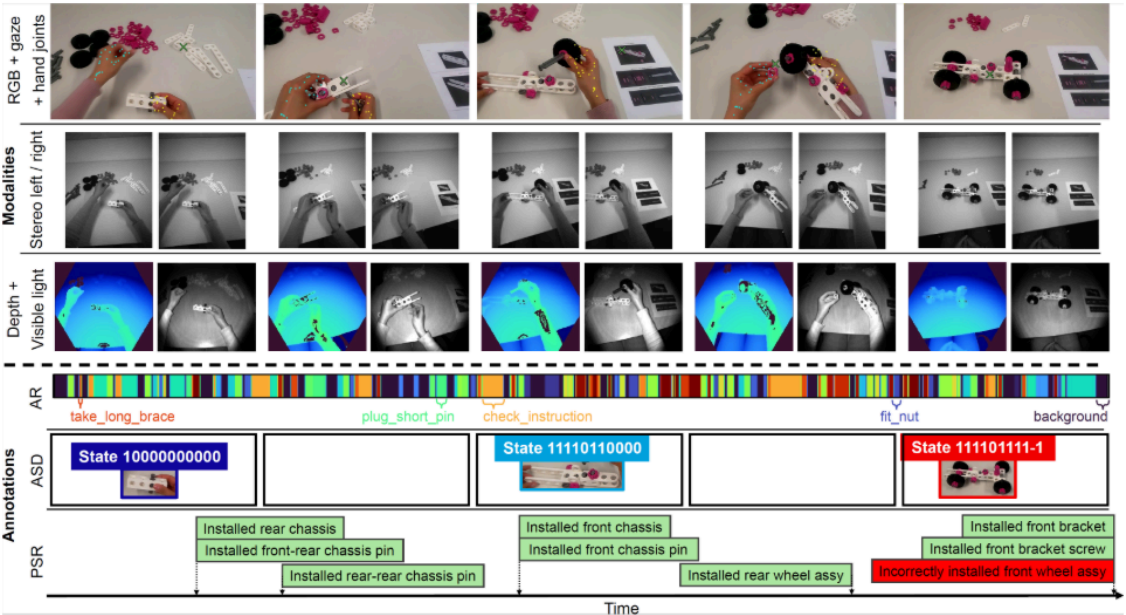


Alternatives Dataset

1. Industreal

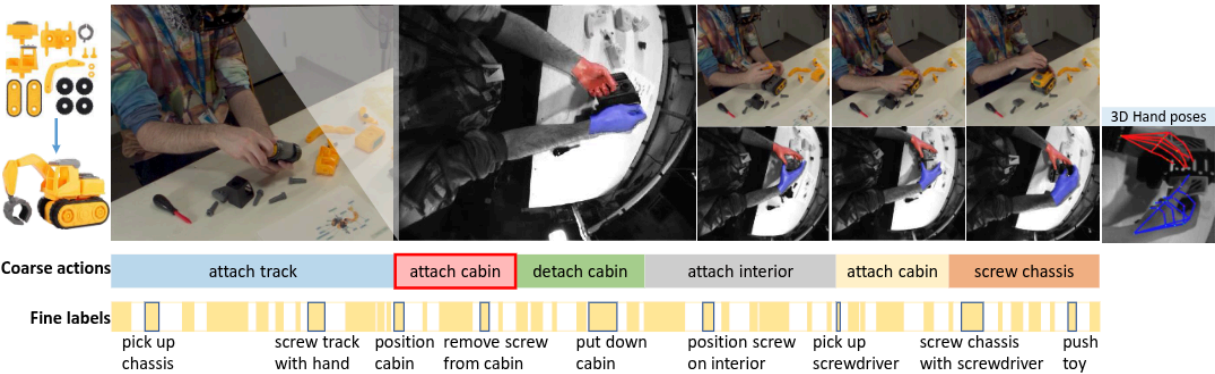
Annotations

IndustReal is specifically created to stimulate research on the novel procedure step recognition task. However, to broaden the possible uses of the dataset, we provide action recognition and object detection (assembly state detection) labels alongside the PSR annotations:



Samples from a clip in the IndustReal dataset, demonstrating the modalities and annotations for all three tasks. Gaze is indicated by the cross, detected hand joints by the dots. AR: action recognition, ASD: assembly state detection, PSR: procedure step recognition.

2. Assembly101



| Dataset     | RGB          | Pose/Keypoints            | Object Detection           | Activity Labels         | Assembly Context     | Size               |
|-------------|--------------|---------------------------|----------------------------|-------------------------|----------------------|--------------------|
| Assembly101 | ✔ Multi-view | ✔ 18M 3D hand poses       | ⚠ Hand only                | ✔ 100K+ coarse, 1M fine | ✔ Toy vehicles       | Large              |
| HA4M        | ✔            | ✔ Skeleton (Azure Kinect) | ✗ Limited                  | ✔ 12 actions            | ✔ Gear train         | Small (217 videos) |
| IndustReal  | ✔ Egocentric | ✔ Hand joints             | ✔ Assembly state detection | ✔ Action + PSR          | ✔ Industrial toy car | Medium (84 videos) |

| Dataset          | RGB                  | Pose/Keypoints     | Object Detection   | Activity Labels      | Assembly Context       | Size       |
|------------------|----------------------|--------------------|--------------------|----------------------|------------------------|------------|
| <b>HA-ViD</b>    | ✓                    | ✗ No pose          | ✓ Object detection | ✓ 219 atomic actions | ✓ Generic assembly box | Medium     |
| <b>Ego-Exo4D</b> | ✓ Ego + third-person | ✓ Body + hand pose | ⚠ Limited          | ✓ Fine-grained       | ✓ Multiple domains     | Very Large |

Priority Schedule:

Now (until the next 2 weeks):

- Implement temporal attention
- Do the ablation training (FiLM and no FiLM, ++)
- Skeleton paper draft

Next 3-4 weeks:

- Assembly101 and Industreal implementation dataset
- gather all of the visualizations needed for the paper

Later 5-6 weeks:

- Benchmarking table
- cross dataset experiment
- Full paper writing